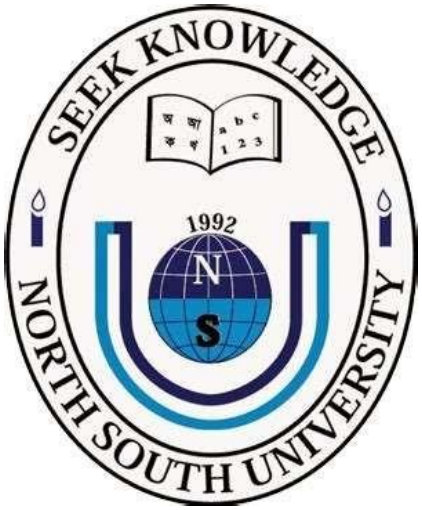
Analog Electronics II: Op-amp & 555 Timer



Course instructor:

Dr. Mohsin Sajjad

Assistant Professor

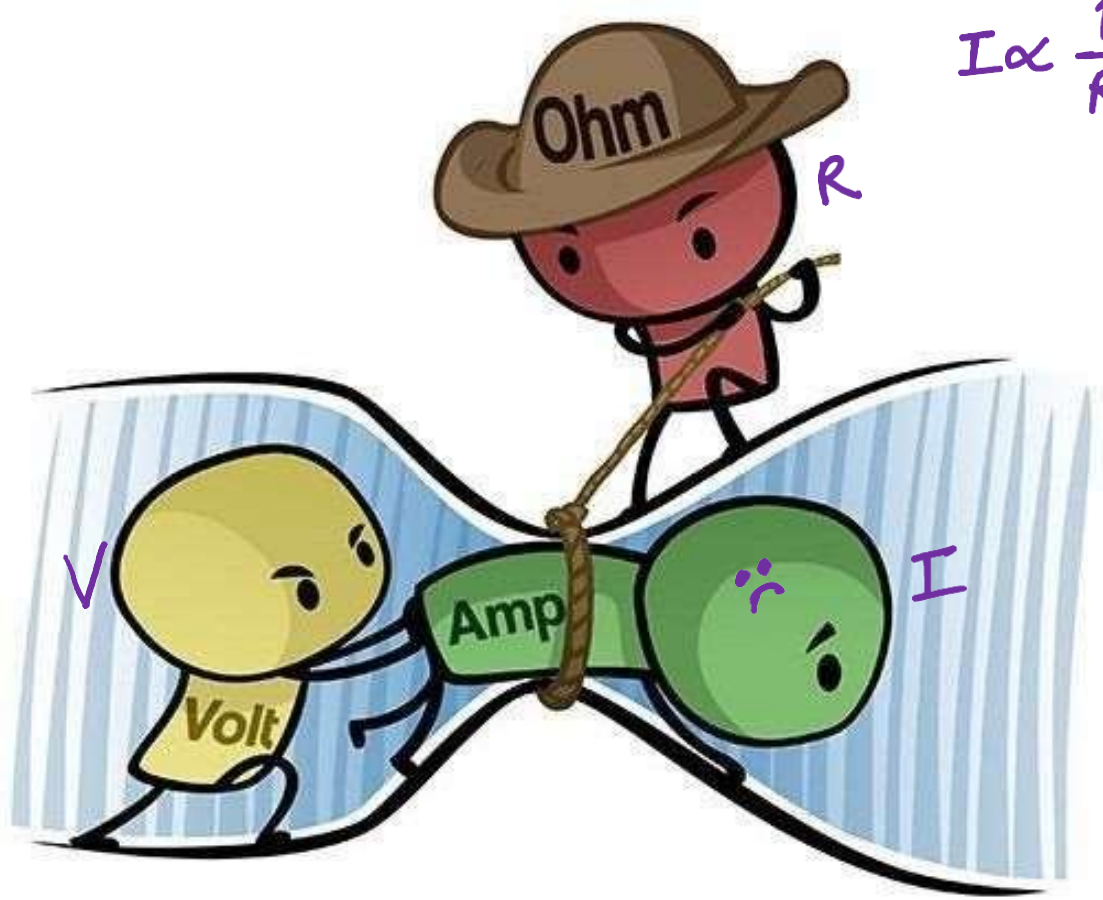
North South University

EEE 311

Lecture # 2

Current, voltage, resistance

Ohm's law



$I \propto \frac{1}{R}$ (V const)

$I \propto V$ (R const)

Source: Amino

Ohm's law

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

That is,

$$v \propto i \tag{2.2}$$

$$\boxed{v = iR}$$

The *resistance* R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

We may deduce from Eq. (2.3) that

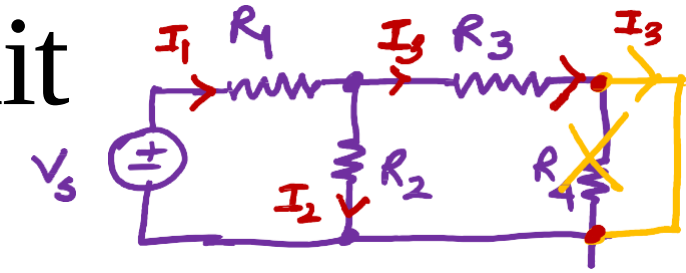
$$R = \frac{v}{i} \tag{2.4}$$

so that

$$1\ \Omega = 1\ \text{V/A}$$

Source: Alexander and Sadiku

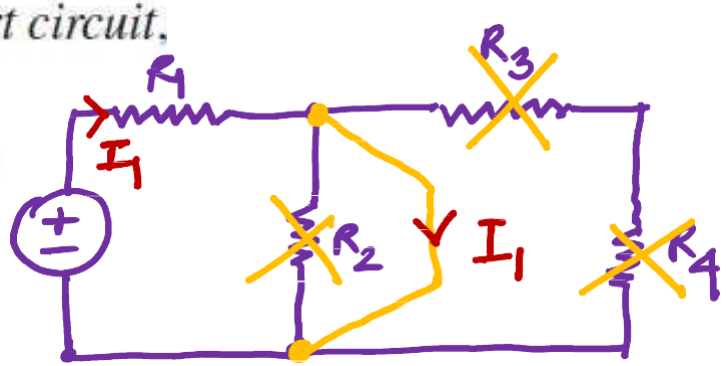
Short circuit and open circuit



A short circuit is a circuit element with resistance approaching zero.

$R = 0$ is called a *short circuit*.

$$v = iR = 0$$



An open circuit is a circuit element with resistance approaching infinity.

$R = \infty$ is known as an *open circuit*.

$$i = \lim_{R \to \infty} \frac{v}{R} = 0$$

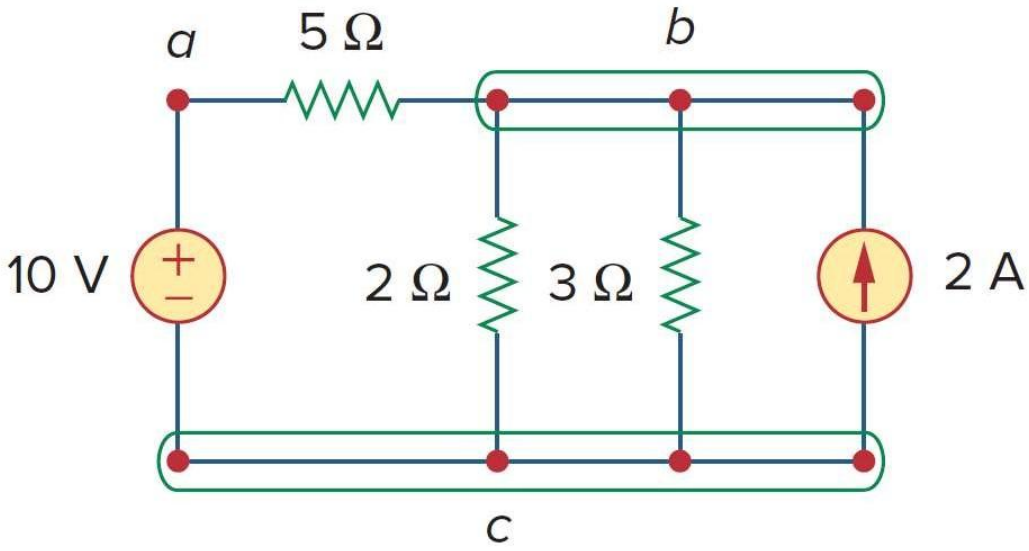
Source: Alexander and Sadiku

Nodes, Branches, and Loops

A branch represents a single element such as a voltage source or a resistor.

A node is the point of connection between two or more branches.

A loop is any closed path in a circuit.



Source: Alexander and Sadiku

Kirchhoff ’s Laws

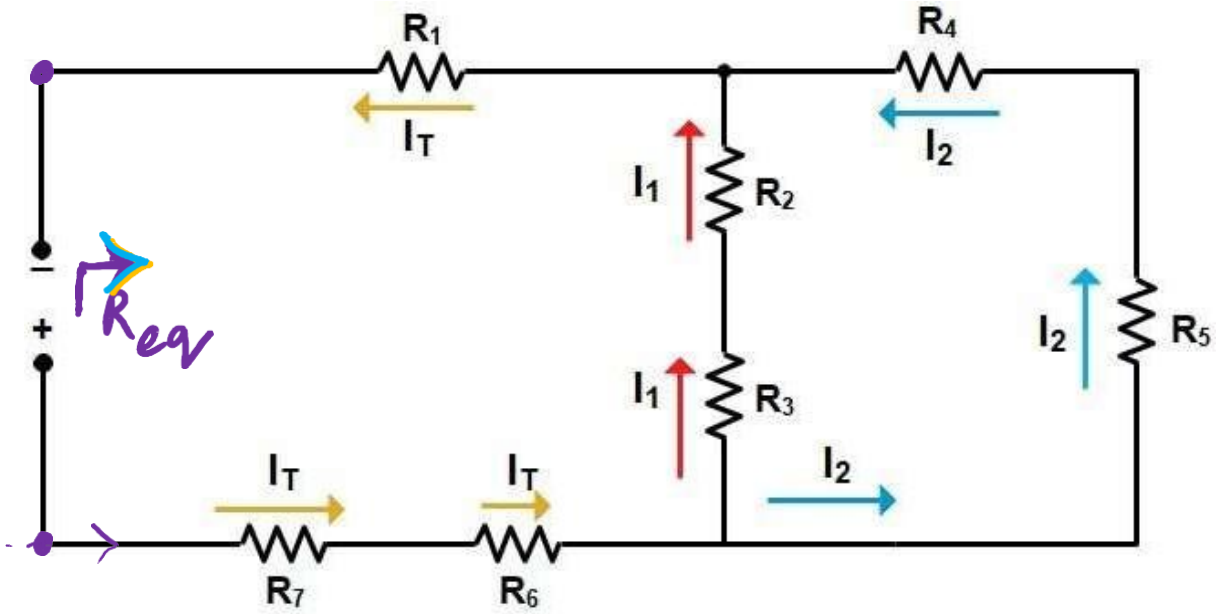
Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

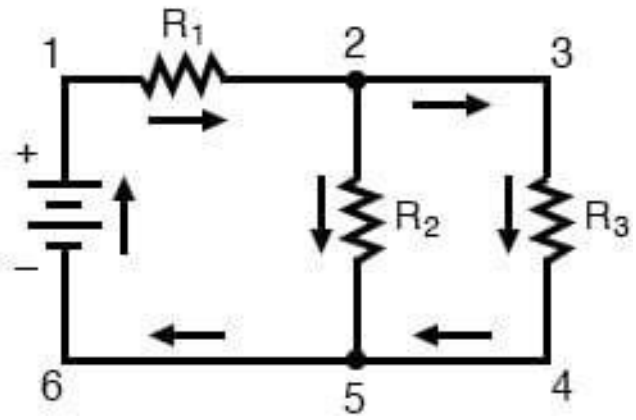
Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

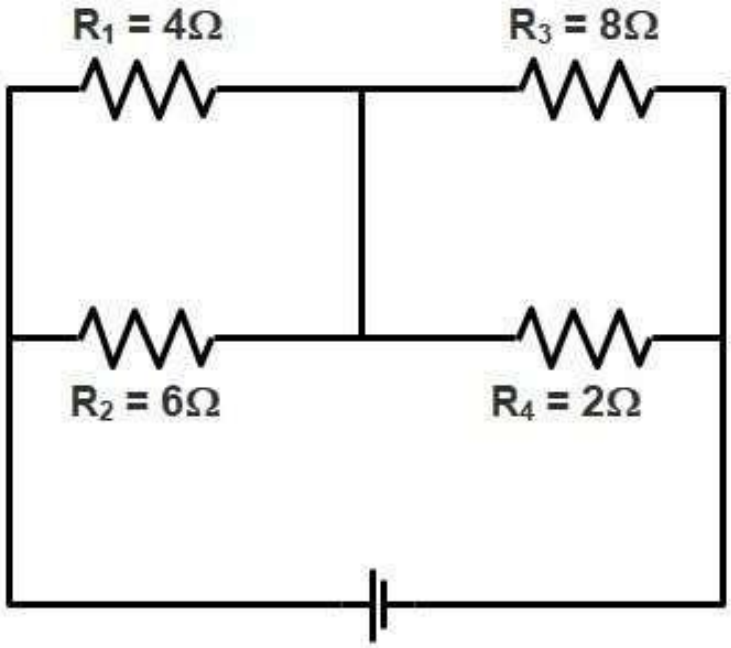
Source: Alexander and Sadiku

Series or parallel or none?



Tracing common current paths to identify series circuit

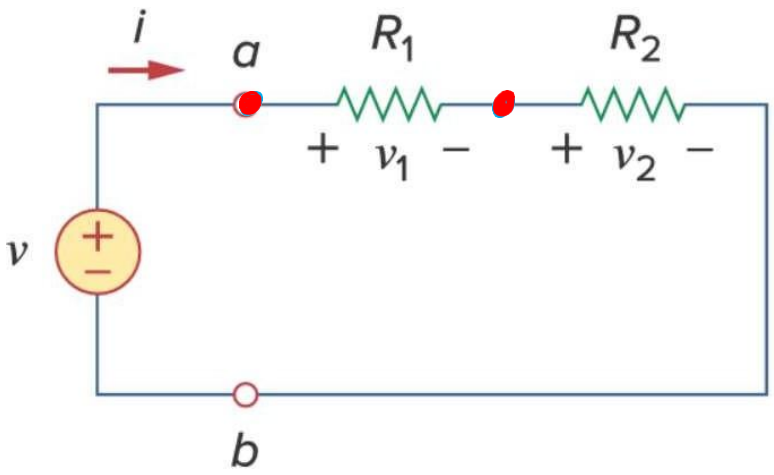


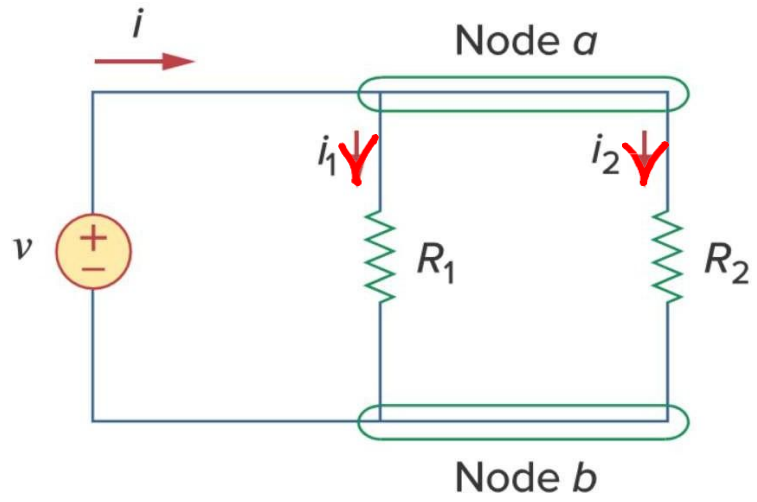


$R_{p_1} + R_{p_2}$

Source: Alexander and Sadiku

Voltage divider rule and Current Divider rule





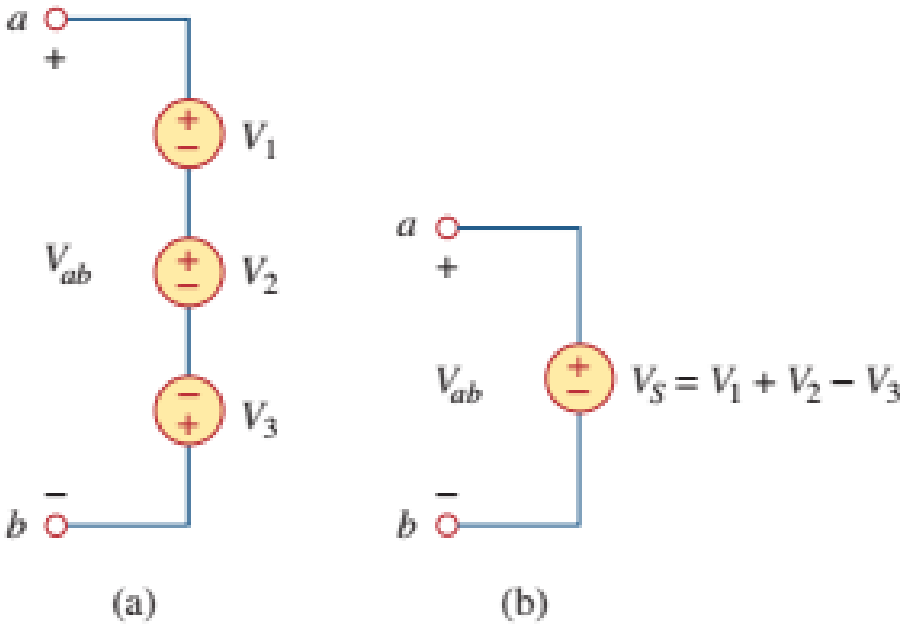
$$v_1 = \frac{R_1}{R_1 + R_2} v$$

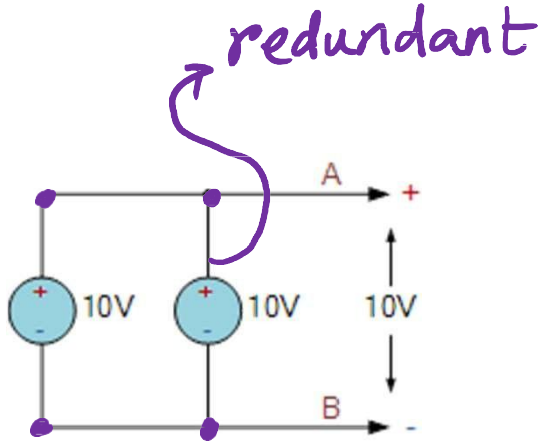
$$v_2 = \frac{R_2}{R_1 + R_2} v$$

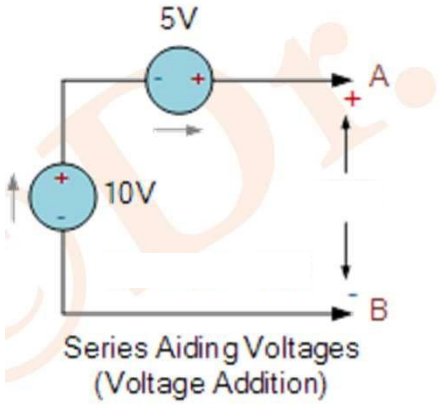
$$i_1 = \frac{R_2\, i}{R_1 + R_2}$$

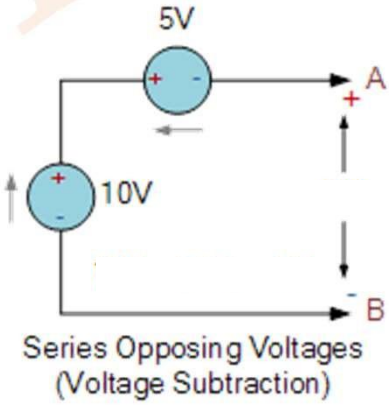
$$i_2 = \frac{R_1\, i}{R_1 + R_2}$$

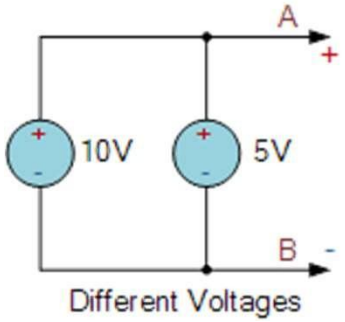
Voltage sources in series and parallel

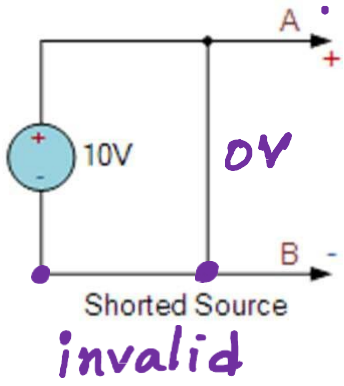






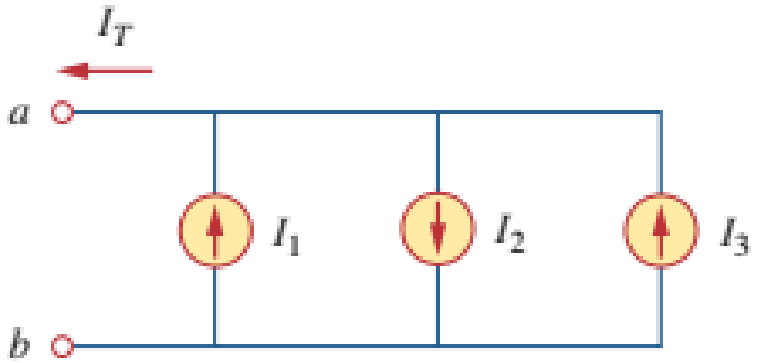




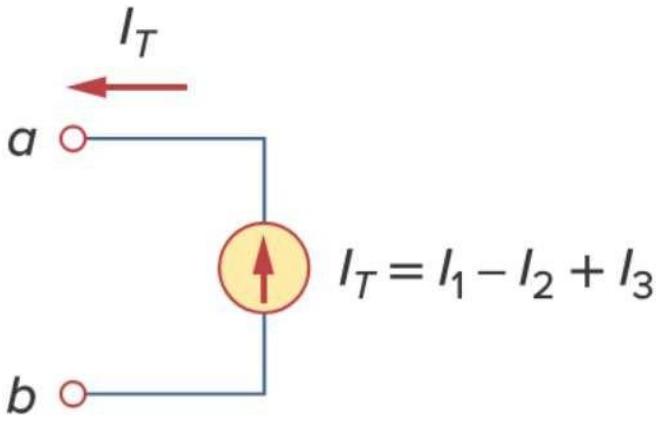


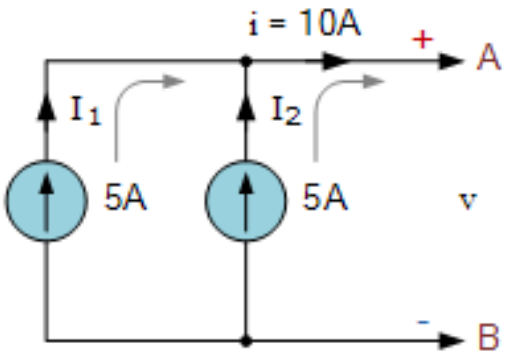
Source: Alexander and Sadiku

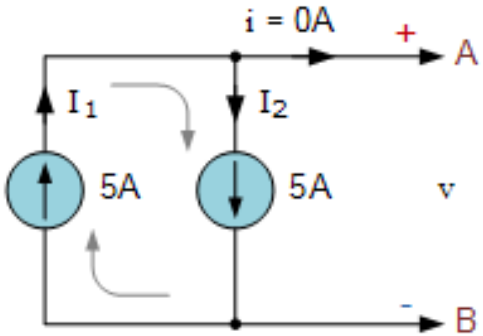
Current sources in series and parallel

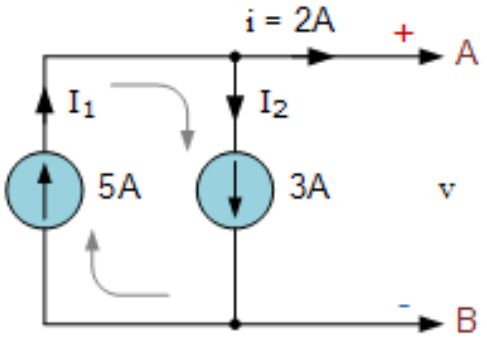


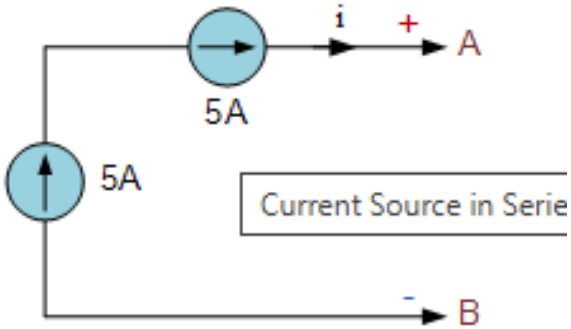
(a)

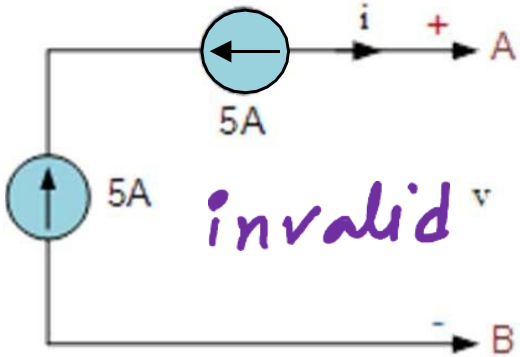










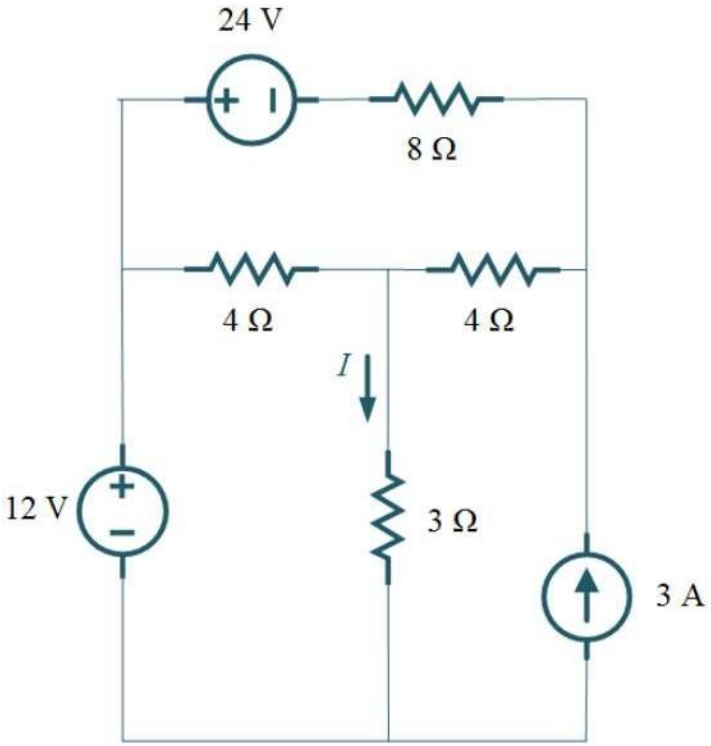


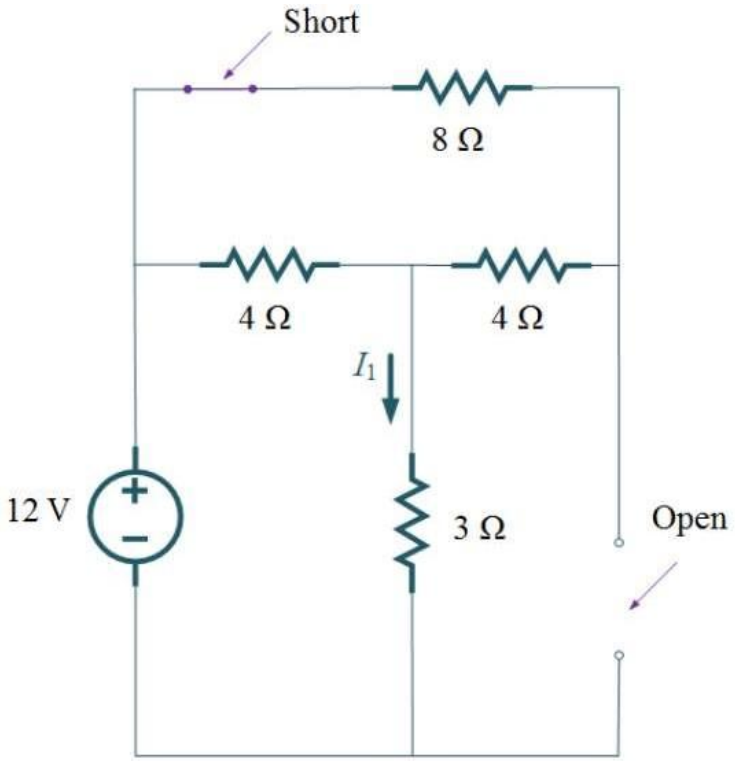
Source: Alexander and Sadiku

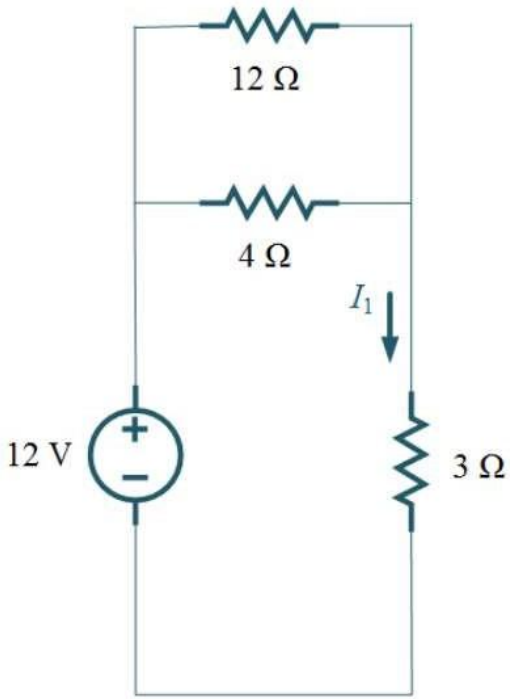
DC circuit problem with multiple sources

- Source transformation theorem
- Superposition principles
- Mesh current analysis
- Node voltage analysis
- Thevenin’s theorem
- Norton’s theorem

Superposition principle



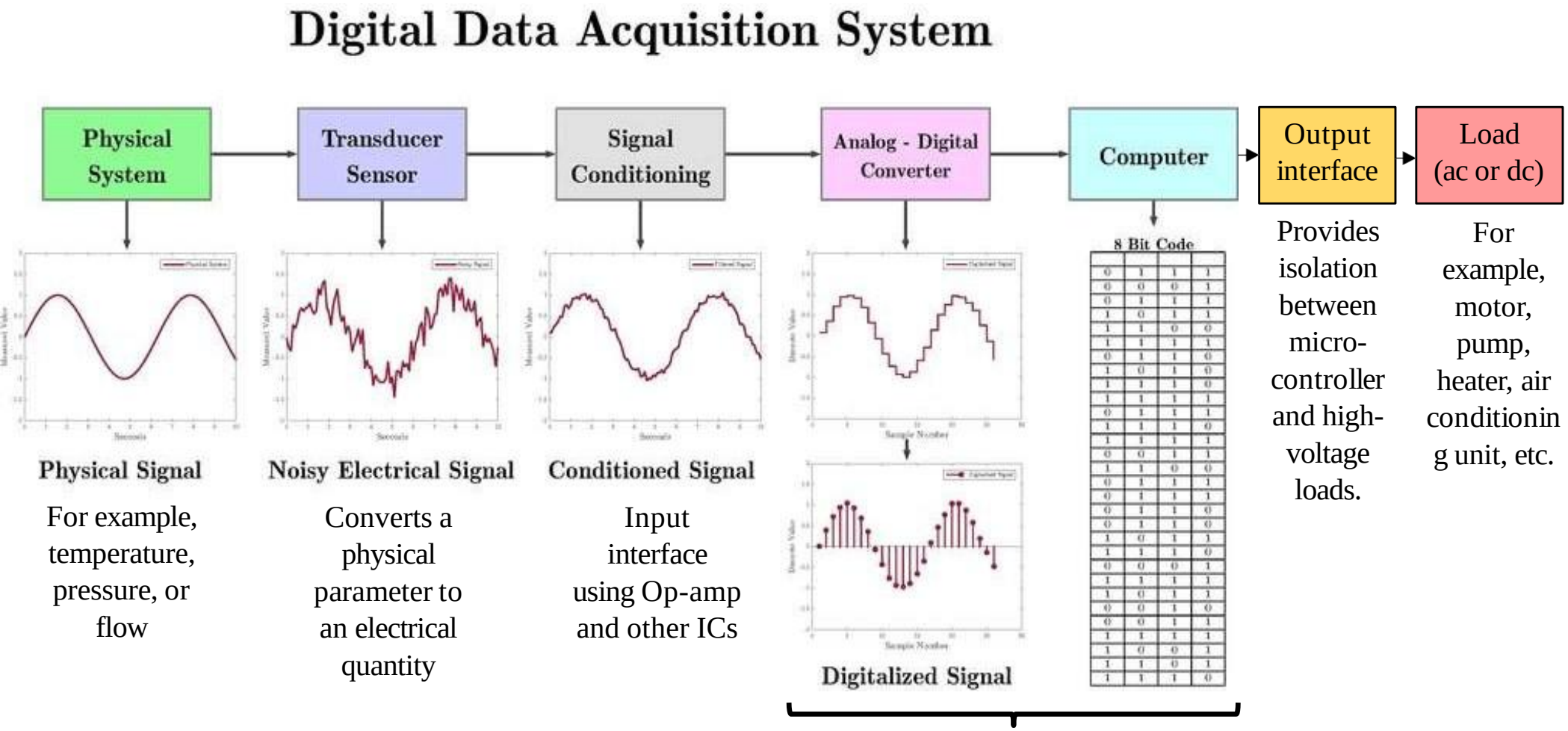




Problem Link: https://electricalworkbook.com/superposition-theorem-in-electric-circuits/

Source: Alexander and Sadiku

Why analog circuit in the digital world?

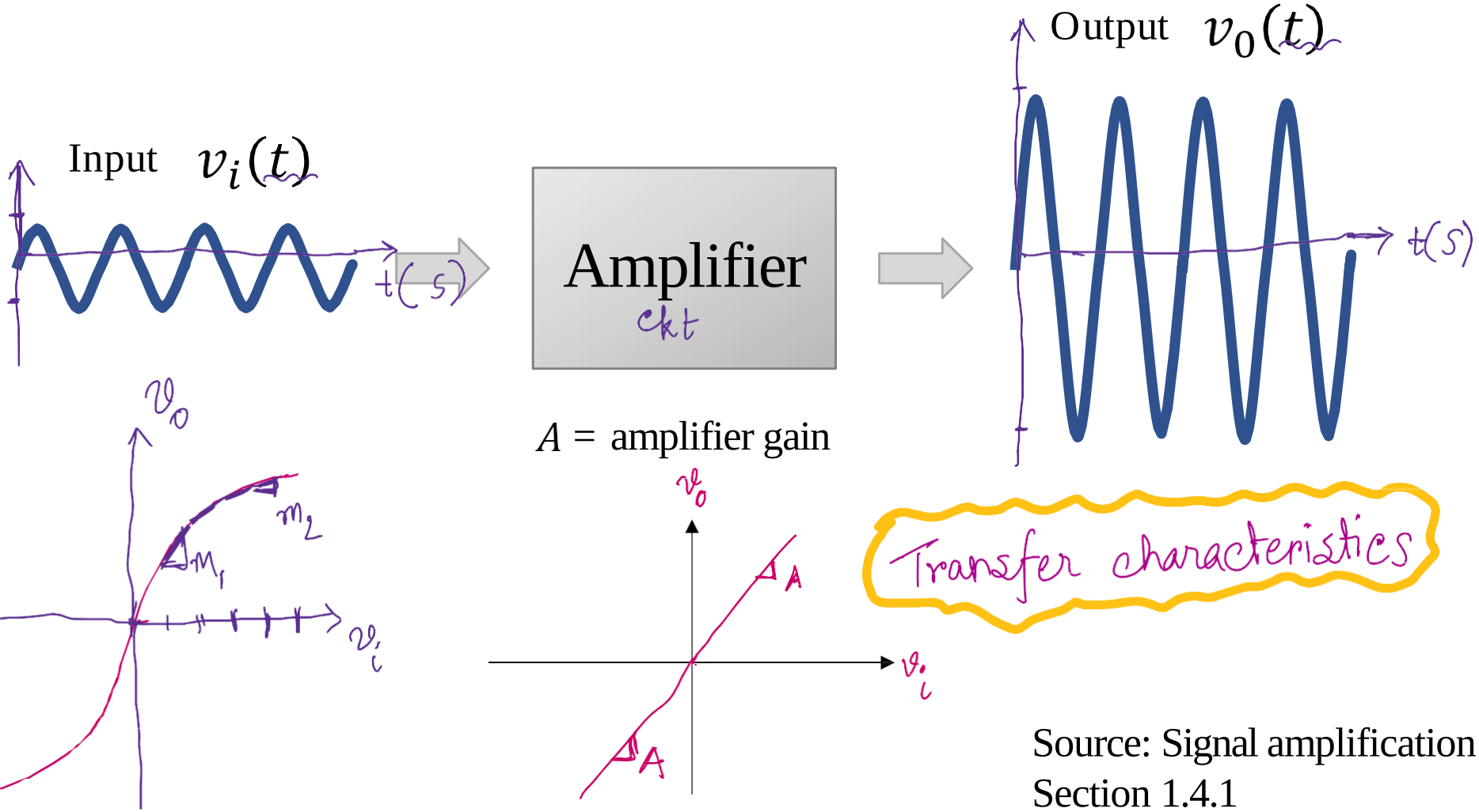


Source: Coughlin, section 1-1 and Wikipedia

Amplifier

For linear amplifier,

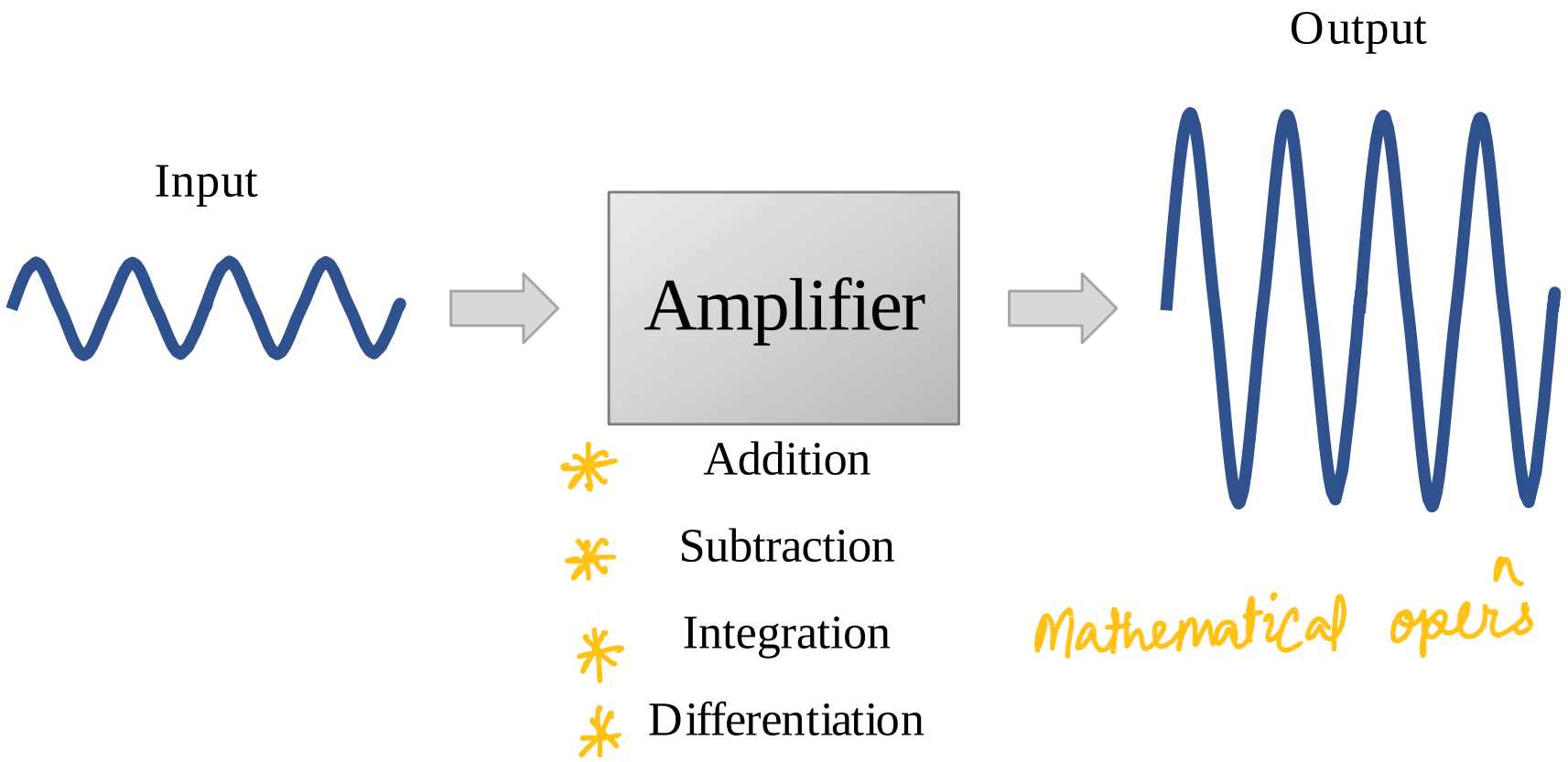
$$v_0(t) = Av_i(t)$$



Source: Signal amplification
Section 1.4.1
Sedra Smith

OP-AMP: an amplifier

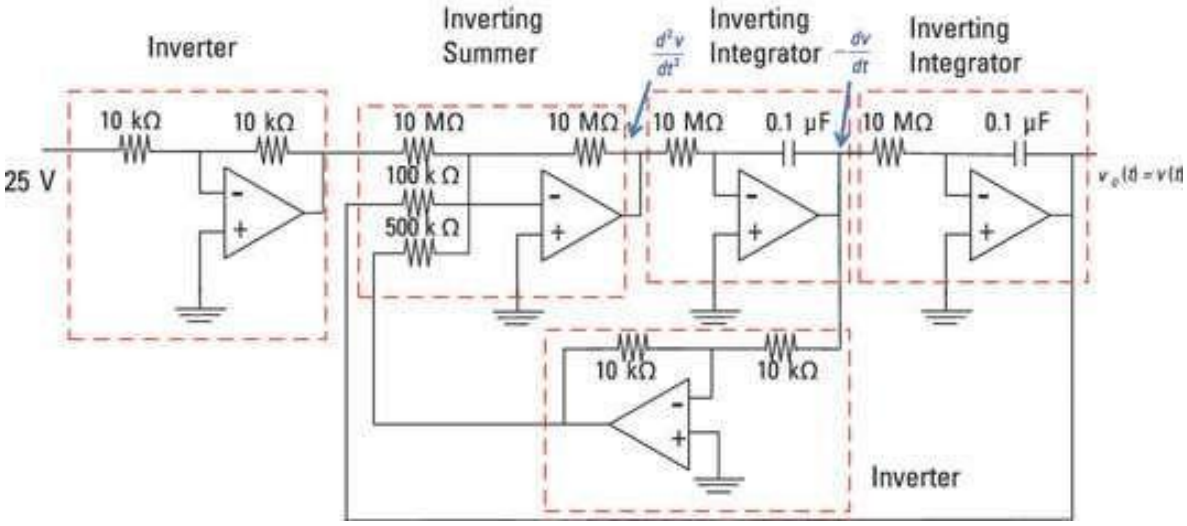
OP-AMP: Operational amplifier



Operational amplifier or Op-Amp

- One of the most versatile and widely used electronic devices in linear applications
- Low-cost and easy-to-use
- Many operations, such as addition, subtraction, multiplication, integration, differentiation, solving differential equations, and so on

$$\frac{d^2v}{dt^2} = -20\frac{dv}{dt} - 100\,v + 25\text{ V}$$



$v_o = v_1 + v_2$

$v_o = v_1 - v_2$

$v_o = 4 \cdot v_1 + 5 \cdot v_2$

$v_o = 2 \cdot v_1 - 8 v_2$

Source: Coughlin, section 1-0

Operational amplifier or Op-Amp

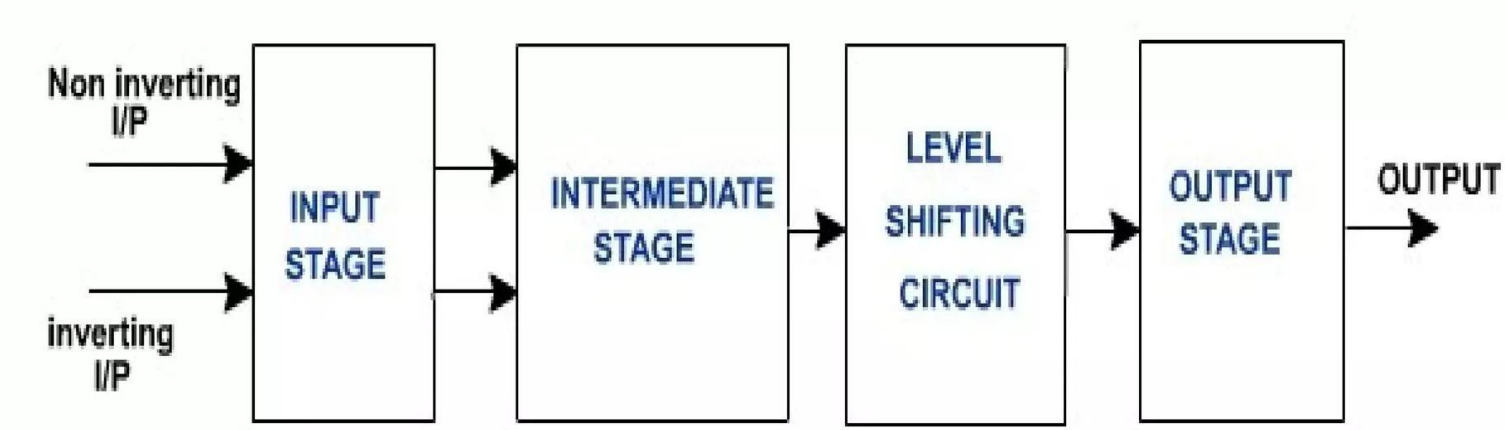
✓An operational amplifier (op-amp) is a DC-coupled high- gain electronic voltage amplifier

✓ Direct- coupled high gain amplifier usually consisting of one or more differential amplifiers

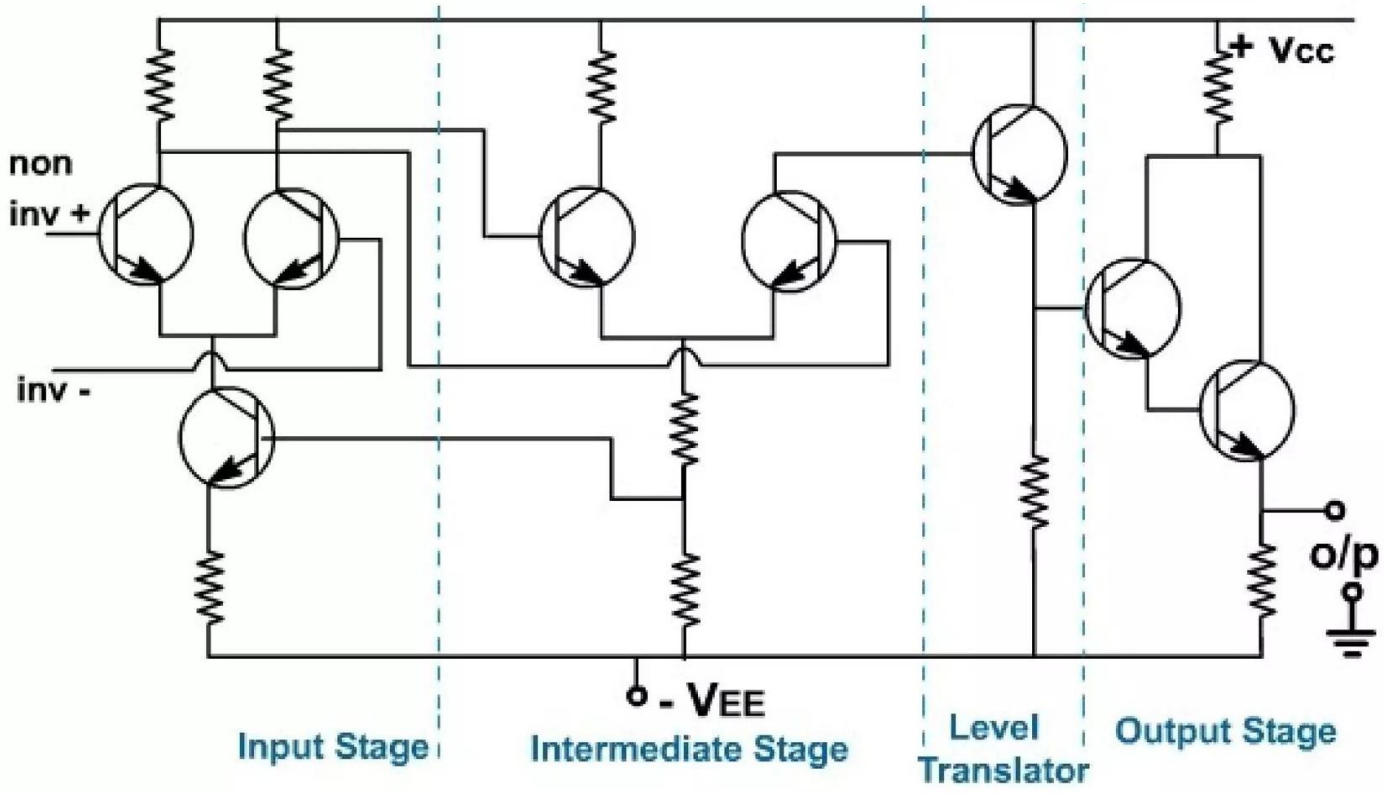
✓Designed for computing mathematical functions such as addition, subtraction ,multiplication, integration & differentiation



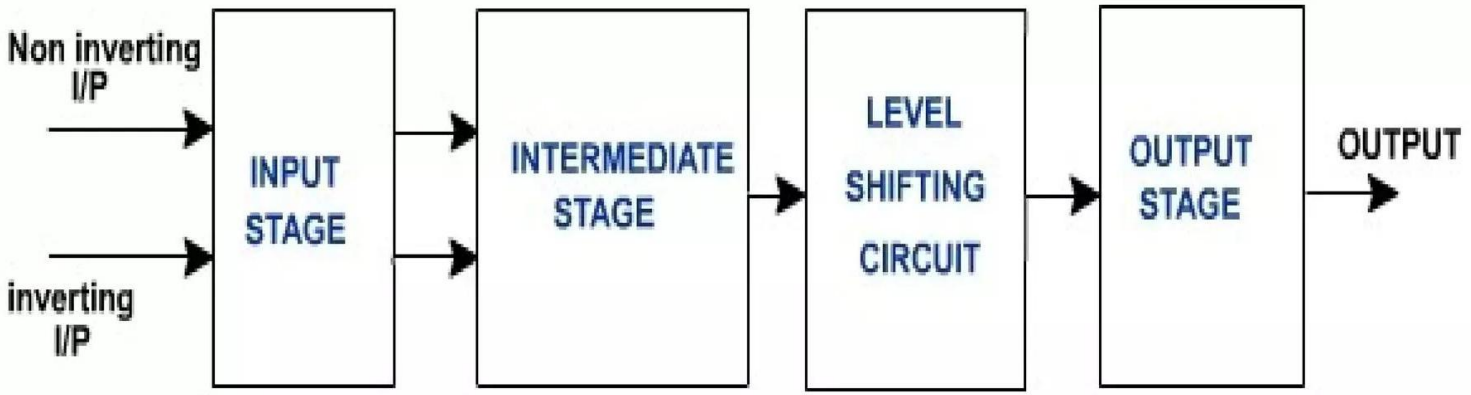
BLOCK DIAGRAM



INTERNAL CIRCUIT DIAGRAM



INTERNAL CIRCUIT DIAGRAM

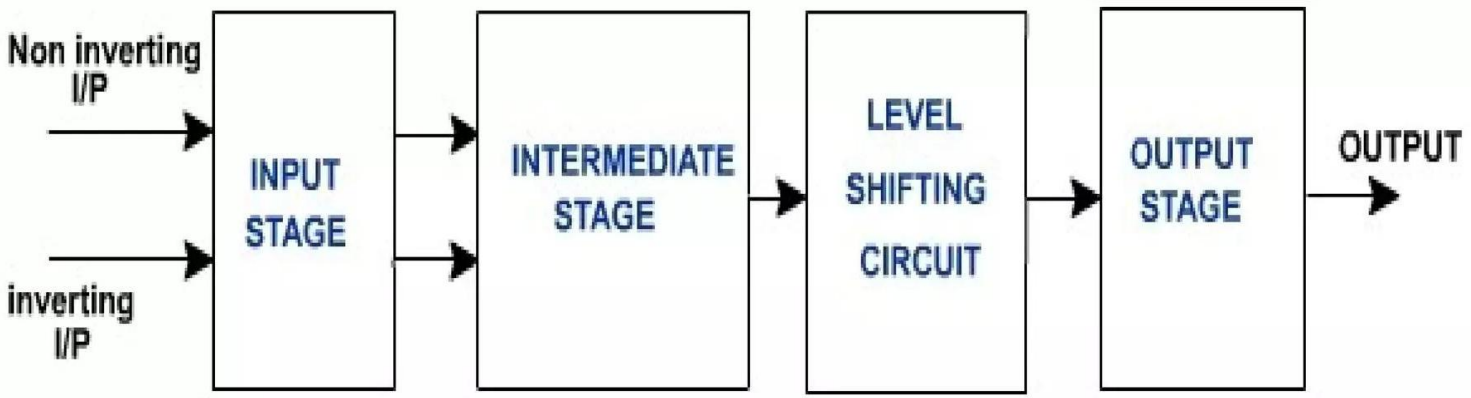


INPUT STAGE

The input stage is a dual input balanced output differential amplifier. This stage provides most of the voltage gain of the amplifier and also establishes the input resistance of the OPAMP.

INTERMEDIATE STAGE

The intermediate stage of OPAMP is another differential amplifier which is driven by the output of the first stage. This is usually dual input unbalanced output.



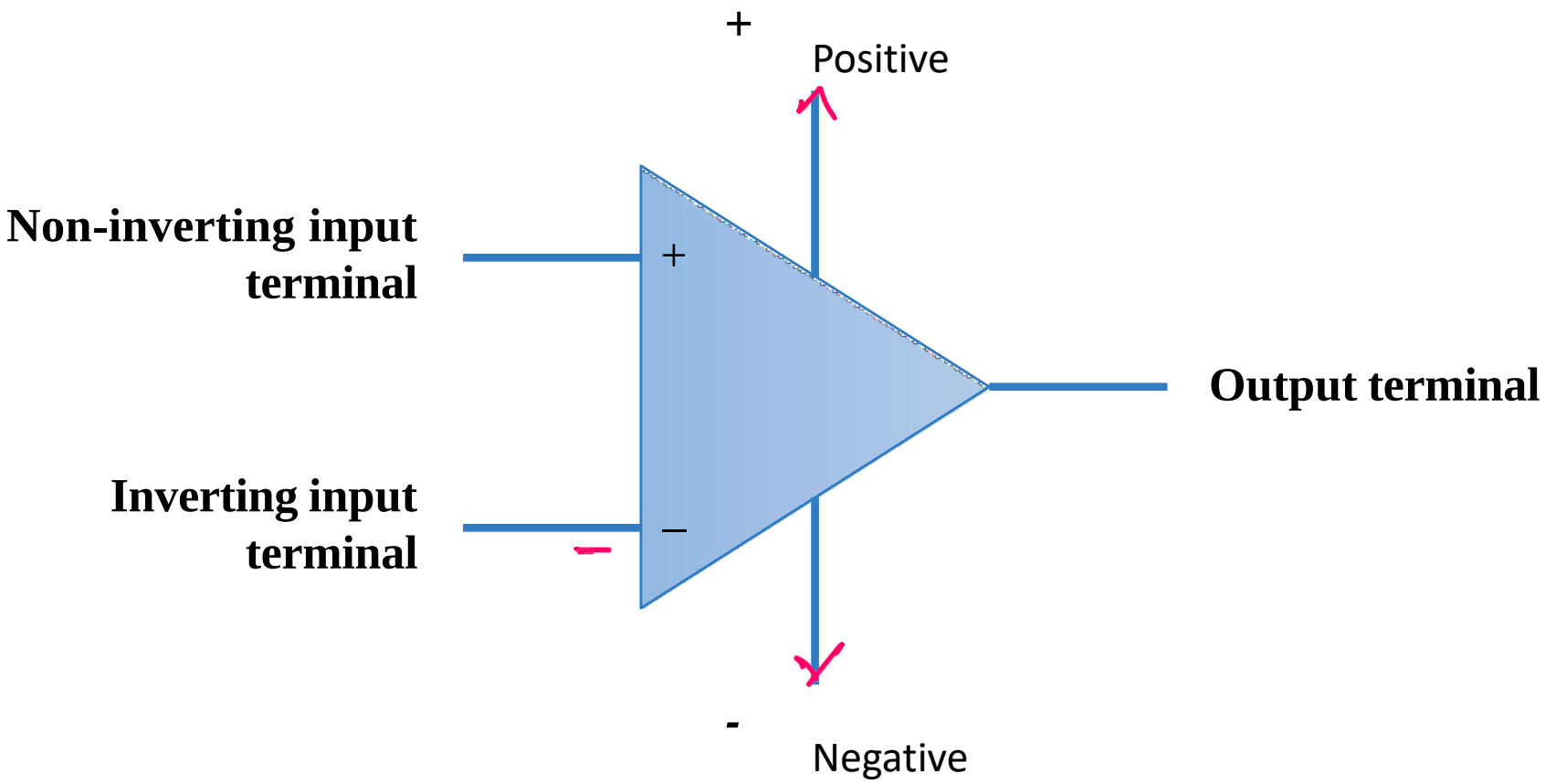
LEVEL SHIFTING STAGE

Because direct coupling is used, the dc voltage level at the output of intermediate stage is well above ground potential. Therefore level shifting circuit is used to shift the dc level at the output downward to zero with respect to ground.

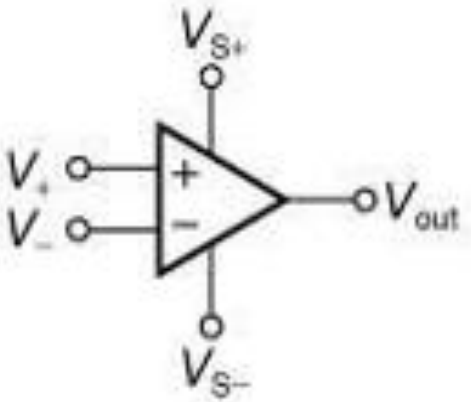
OUTPUT STAGE

The output stage is generally a push pull complementary amplifier. The output stage increases the output voltage swing and raises the current supplying capability of the OPAMP. It also provides low output resistance.

OP-AMP circuit symbol



Circuit symbol

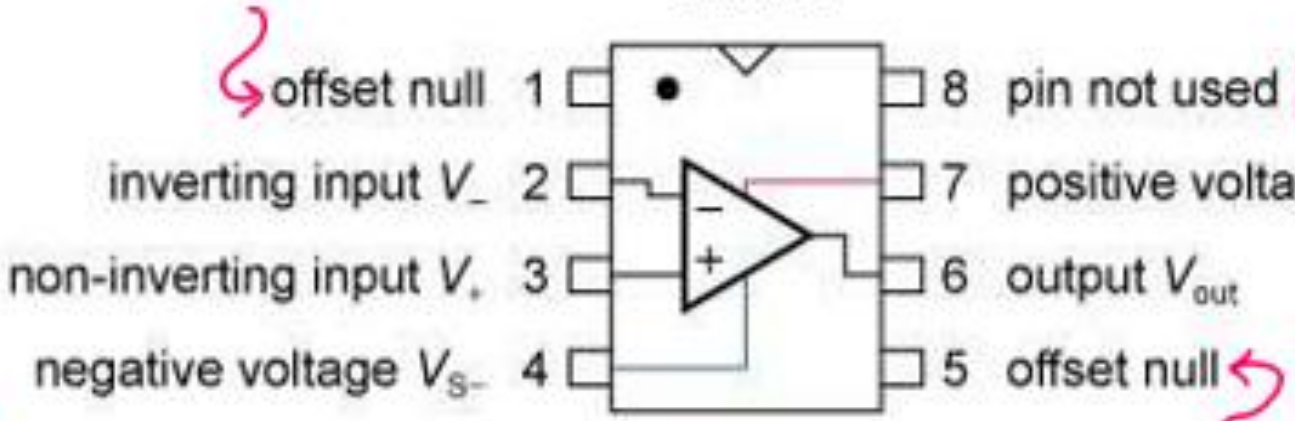


(a)

Dual-in-line package (DIP)



(b) Terminals are numbered counter-clockwise

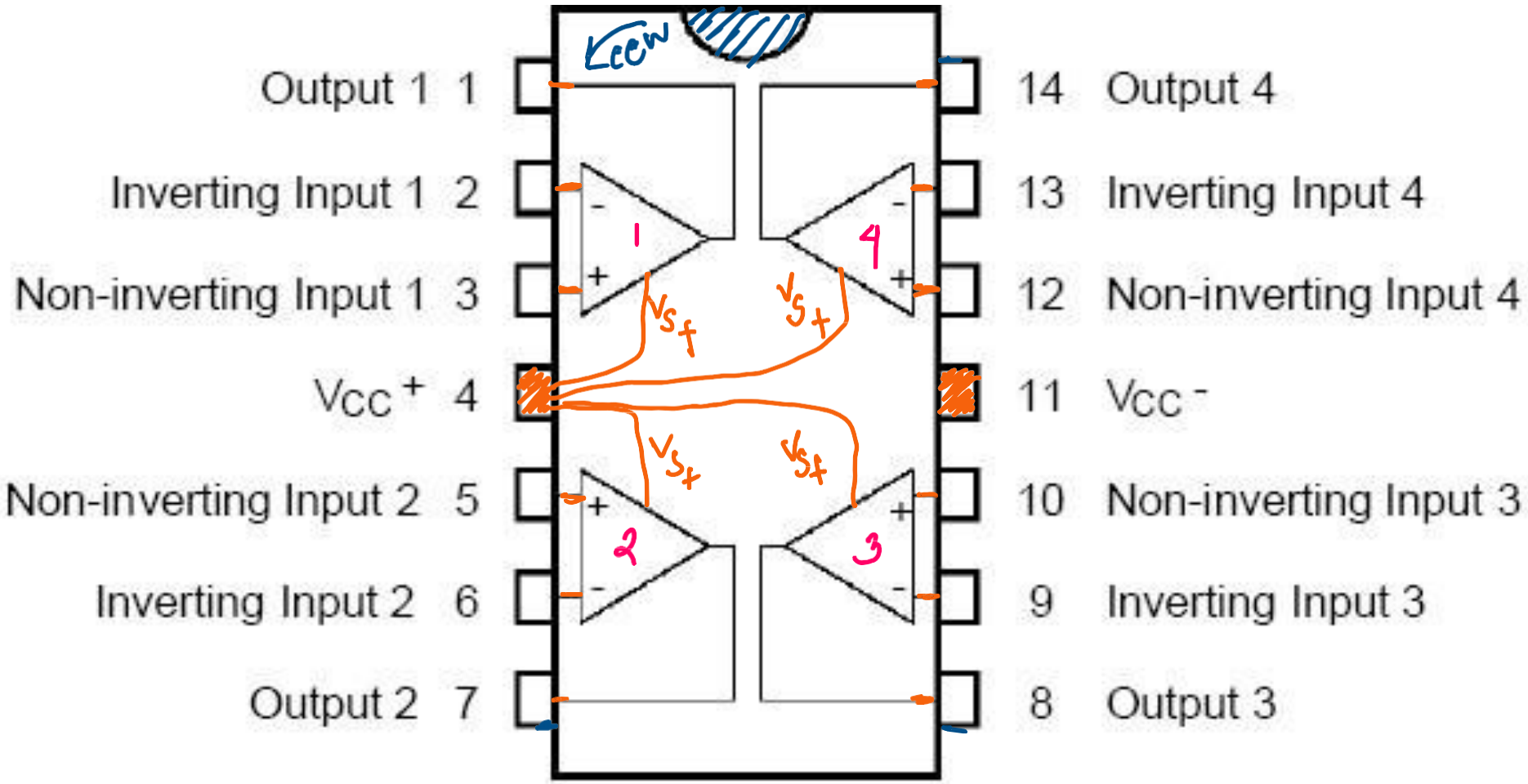


Connection diagram

(c)

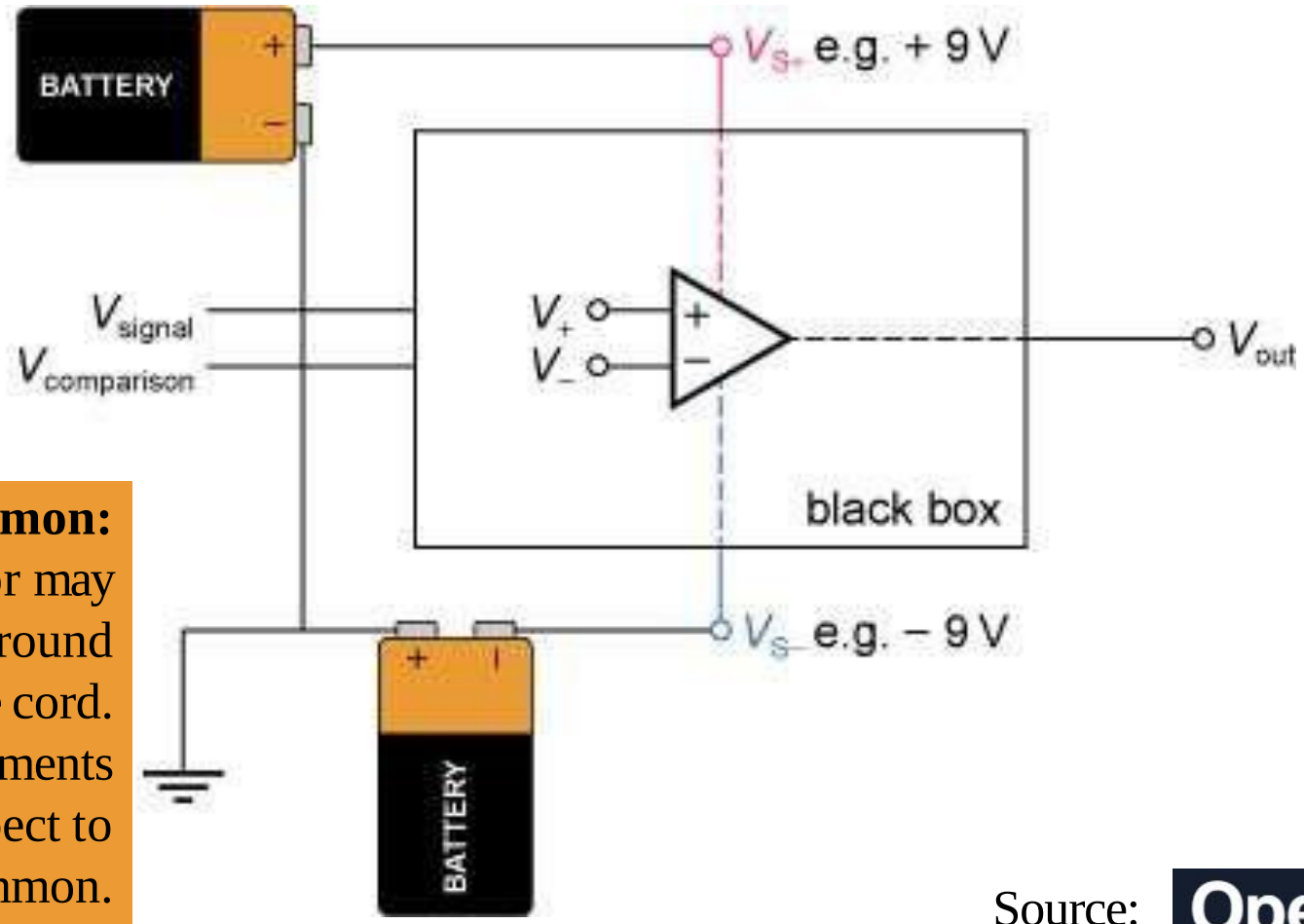
Source: OpenLearn

Quad op-amp



Source: Soundtronics.co.uk

Dual power supplies



Power supply common: This terminal may or may not be wired to earth ground via the 3rd wire of line cord. All voltage measurements are made with respect to power supply common.

Source: OpenLearn

Coughlin 1-6.1, 2-1.1